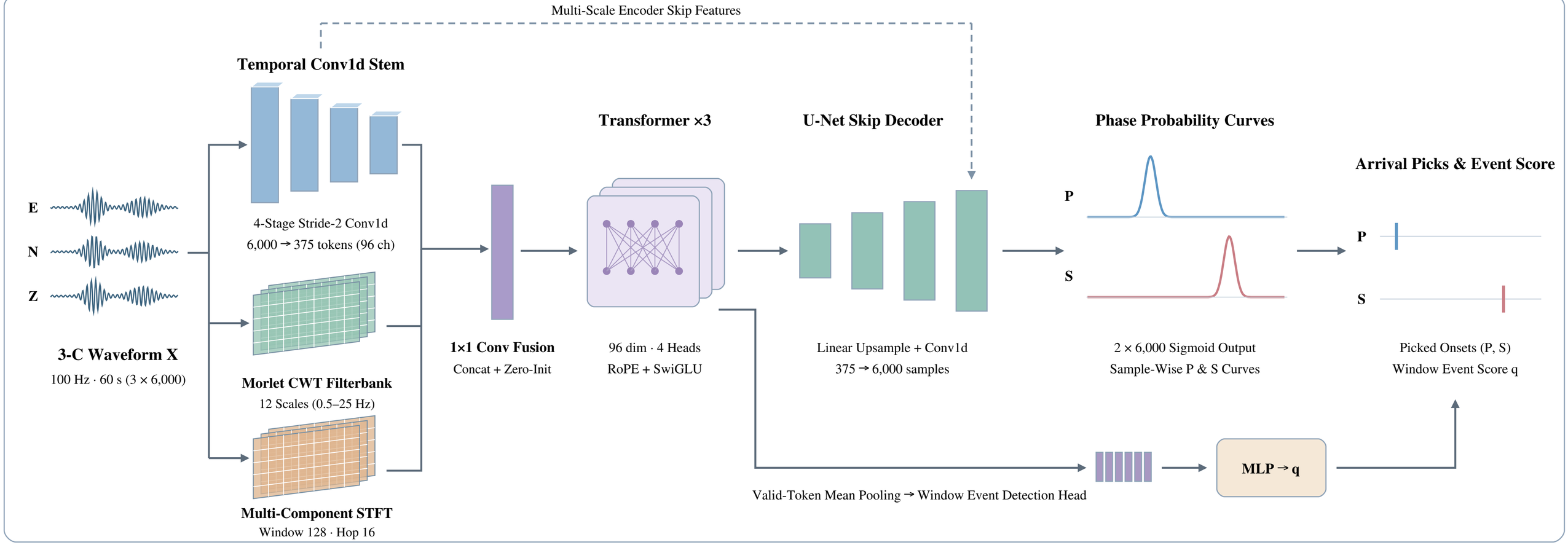

Multi-Scale Encoder Skip Features
Temporal Conv1d Stem
E
N
Z
3-C Waveform X
100 Hz · 60 s (3 × 6,000)
4-Stage Stride-2 Conv1d
6,000 → 375 tokens (96 ch)
Morlet CWT Filterbank
12 Scales (0.5–25 Hz)
Multi-Component STFT
Window 128 · Hop 16
1×1 Conv Fusion
Concat + Zero-Init
Transformer ×3
96 dim · 4 Heads
RoPE + SwiGLU
U-Net Skip Decoder
Linear Upsample + Conv1d
375 → 6,000 samples
Phase Probability Curves
P
S
2 × 6,000 Sigmoid Output
Sample-Wise P & S Curves
Arrival Picks & Event Score
P
S
Picked Onsets (P, S)
Window Event Score q
Valid-Token Mean Pooling → Window Event Detection Head
MLP → q